MIRMAHDI SEYEDREZAEI

Ph.D. Candidate, University of Southern California
iLAB, 3620 S. Vermont Avenue, Kaprielian Hall 217 & 224C, Los Angeles, CA 90089-2531
seyedrez@usc.edu | +1 (213) 667-7404 | LinkedIn | Google Scholar

EDUCATION

Ph.D. in Civil Engineering, University of Southern California (Expected) 2026
Advisor: Professor Burcin Becerik-Gerber

M.S. in Computer Science, University of Southern California 2026

M.S. in Sustainable Engineering, University of Southern California 2025

M.S. in Civil Engineering (Earthquake Engineering), Sharif University of Technology, Tehran, Iran 2018
Thesis: Structural Health Monitoring Considering Mass and Stiffness Modification

B.S. in Civil Engineering, Sharif University of Technology, Tehran, Iran 2016

RESEARCH INTERESTS

My research investigates how built environments shape human health, safety, and well-being, and how intelligent systems can be designed to make that relationship adaptive rather than indifferent. I develop human-centered cyber-physical systems that integrate environmental sensing, wearable physiology, and machine learning to understand how indoor conditions, energy systems, and infrastructure performance affect people, from hybrid workplaces and affordable housing to construction operations. This foundation extends into construction informatics: my current work uses BIM-encoded structural priors within graph neural networks and transformer architectures to forecast building automation system behavior, detect design-to-operations drift, and establish data governance and privacy standards for building digital twins, bringing the same occupant-centered, data-driven approach to how buildings are monitored and operated. Across these threads, my methods span controlled experiments, longitudinal field deployments, and machine learning, always oriented toward rigorous, deployable solutions for digital engineering practice.

PUBLICATIONS

Journal Publications (Published)

1. **M. Seyedrezaei**, B. Becerik-Gerber, S. C. Roll, G. Lucas, and S. Narayanan. "How Environmental Conditions Shape Workload in Hybrid Work: A Longitudinal Mediation Analysis." *Building and Environment*, p. 114973, 2026. doi:10.1016/j.buildenv.2026.114973
2. **M. Seyedrezaei**, B. Becerik-Gerber, and T. J. Kohut. "Exploring energy consumption in energy-efficient, affordable housing in Southern California: Behavioral insights and financial implications." *Energy and Buildings*, 331:115369, 2025. doi:10.1016/j.enbuild.2025.115369
3. **M. Seyedrezaei**, M. Awada, S. Contreras, B. Becerik-Gerber, and G. Boeing. "Equity in the built environment: A systematic review." *Building and Environment*, 245:110827, 2023. doi:10.1016/j.buildenv.2023.110827
4. **M. Seyedrezaei**, A. N. Tak, and B. Becerik-Gerber. "Consumption and conservation behaviors among affordable housing residents in Southern California." *Energy and Buildings*, 304:113840, 2024. doi:10.1016/j.enbuild.2023.113840
5. **M. Seyedrezaei**, M. Awada, B. Becerik-Gerber, G. Lucas, and S. Roll. "Interaction effects of indoor environmental quality factors on cognitive performance and perceived comfort of young adults in open-plan offices in a North American Mediterranean climate." *Building and Environment*, 244:110743, 2023. doi:10.1016/j.buildenv.2023.110743
6. M. Awada, B. Becerik-Gerber, R. Liu, **M. Seyedrezaei**, et al. "Ten questions concerning the impact of environmental stress on office workers." *Building and Environment*, 229:109964, 2023. doi:10.1016/j.buildenv.2022.109964
7. R. Moaddab, K. Amini Hosseini, and **M. Seyedrezaei**. "A holistic approach for evaluating the system performance and earthquake resilience in historic-commercial urban fabrics." *Natural Hazards*, 116(1):1261–1289, 2023. doi:10.1007/s11069-022-05720-1
8. B. Becerik-Gerber, G. Lucas, A. Aryal, M. Awada, et al. (incl. **M. Seyedrezaei**). "The field of human building interaction for convergent research and innovation for intelligent built environments." *Scientific Reports*, 12(1):22092, 2022. doi:10.1038/s41598-022-25047-y
9. B. Becerik-Gerber, G. Lucas, A. Aryal, M. Awada, et al. (incl. **M. Seyedrezaei**). "Ten questions concerning human-building interaction research for improving the quality of life." *Building and Environment*, 226:109681, 2022. doi:10.1016/j.buildenv.2022.109681

Journal Publications (Accepted / In Press)

10. **M. Seyedrezaei**, B. Becerik-Gerber, S. C. Roll, G. Lucas, and S. Narayanan. "Longitudinal Exploration of Productivity and Well-being in Hybrid Work Through Ecological Momentary Assessment and Environmental Sensing." *Journal of Occupational and Environmental Medicine*, accepted, 2026.
11. **M. Seyedrezaei**, B. Becerik-Gerber, M. Awada, G. Lucas, and S. Roll. "Interaction effects of indoor environmental quality on stress in offices." *Indoor and Built Environment*, accepted, 2026.

Journal Publications (Submitted)

12. M. Reyes, T. Zang, **M. Seyedrezaei**, R. Liu, B. Becerik-Gerber, G. M. Lucas, S. C. Roll, and S. Narayanan. "Sleep-work dynamics in naturalistic settings: A four-month longitudinal study." Submitted to *Sleep Health*, 2026.
13. R. Tajik, N. Varman, **M. Seyedrezaei**, M. Awada, B. Becerik-Gerber, et al. "Charting the Future of Occupant-Centric Enabling Technologies for Well-Being in Buildings: Challenges, Gaps, and a Research and Application Roadmap." Submitted to *Building Simulation*, 2026.

Journal Publications (Under Preparation)

14. **M. Seyedrezaei**, B. Becerik-Gerber, S. Roll, G. Lucas, and S. Narayanan. "Multi-scale regimes of indoor environmental quality and their links to occupant well-being." In preparation, 2026.
15. **M. Seyedrezaei**, B. Becerik-Gerber, S. Roll, G. Lucas, and S. Narayanan. "Privacy-utility trade-offs and re-identification risk in behavioral state sequences derived from indoor environmental sensing." In preparation, 2026.

Conference Publications (Published)

16. **M. Seyedrezaei**, M. Reyes, B. Becerik-Gerber, S. C. Roll, G. Lucas, and S. Narayanan. "Longitudinal Analysis of IEQ Variability in Office and Home Workspaces: Insights for the Future of Hybrid Work." *Proc. ASCE Int. Conf. on Computing in Civil Engineering*, pp. 1058–1067, 2026. doi:10.1061/9780784486436.113
17. **M. Seyedrezaei** and B. Becerik-Gerber. "Occupant Satisfaction with Building Systems and Thermostat Usage in Energy-Efficient Affordable Housing in Southern California: Preliminary Findings." *CIB Conferences*, 1:174, 2025. doi:10.7771/3067-4883.1929
18. **M. Seyedrezaei**, B. Becerik-Gerber, S. C. Roll, and G. Lucas. "A Preliminary Investigation for Understanding Temperature's Associations with Office Workers' Psychological Well-Being: A Close Look into Anxiety, Motivation, and Concentration." *Proc. ASCE Int. Conf. on Computing in Civil Engineering*, pp. 568–572, 2025. doi:10.1061/9780784486139.061
19. M. Awada, **M. Seyedrezaei**, B. Becerik-Gerber, and G. Lucas. "Investigating the Interplay Between Indoor Environmental Quality and Workers' Health and Productivity: Preliminary Results." *Proc. ASCE Int. Conf. on Computing in Civil Engineering*, pp. 614–622, 2024. doi:10.1061/9780784485248.074

Conference Publications (Accepted)

20. **M. Seyedrezaei**, B. Becerik-Gerber, S. C. Roll, G. Lucas, and S. Narayanan. "Daily IAQ and Performance in Hybrid Work: A Longitudinal Comparison of Home and Office." *Proc. Indoor Air 2026*, accepted.

Books

21. **M. Seyedrezaei**. *Iran Nationwide Flood*. Lohe Fekr Publishing Agency, 2024. (Edited conference proceedings, Sharif University of Technology, 2019.)

RESEARCH EXPERIENCE

Behavioral Inference, Privacy Governance, and BIM-Informed Operational Intelligence in Building Digital Twins/Shadows 2026–Present

Co-Lead Researcher

- Leading observational studies using the GCS building digital twin/shadow at USC, a read-only cyber-physical platform integrating 252 BAS-connected devices, 1,000+ sensors across seven floors, and matched architectural, mechanical, and structural BIM files
- Developing a five-level HVAC-native behavioral inference hierarchy and quantifying re-identification risk across BAS telemetry; proposing the BIM privacy floor concept and designing a revealed-preference human subjects study (n=60–80) to establish empirical foundations for building data governance standards
- Implementing and comparing four model classes (graph neural networks, transformers, Gaussian processes, and regularized linear models) with and without BIM-encoded topology as a structural prior, to improve zone-level thermal/airflow forecasting and enable cross-building generalization
- Developing a zone-level performance gap index linking BIM geometric features to operational failure patterns, and a topology inference method to detect post-commissioning drift between design and as-operated HVAC configu-

rations
- Mentoring Yuxin Yang (Ph.D. student) on datasets, research questions, and analytical approach for the behavioral inference work
- Collaborating with Dr. David Gerber (lead of the GCS digital shadow platform), Dr. Arsalan Heydarian (UVA), and Drs. Gale Lucas and Burcin Becerik-Gerber on proposal development, including an ongoing NSF SaTC privacy proposal

In preparation for NSF SaTC 2.0 (NSF 25-515), CEC EPIC 5 (GFO-25-308)

Dynamic Workstyles and Worker Health: A Continuous Feasibility Study 2026–Present
Principal Investigator-Lead Researcher

- Authored the funded research proposal independently, designed the study protocol, and instruments
- Designed a four-device sensor suite protocol (physiological, lighting, noise, air quality sensing) paired with ecological momentary assessments to introduce person-centered, location-agnostic continuous monitoring and explicit stress source-attribution as new methodological contributions to occupational health research
- Leading participant recruitment, data collection, and analysis for a two-week continuous monitoring study of 20 office-based knowledge workers
- Mentoring Tao Zang (Ph.D. student) on study tools, device selection, and study design
- Collaborating with Professors Burcin Becerik-Gerber, Gale Lucas, and Shawn Roll on proposal development and study design, and reporting on study progress to Professor Jian Li, M.D., Ph.D., Professor of Work and Health at UCLA's Fielding School of Public Health, as SCERC Program Director, and to the broader SCERC program directorship

Funded by the Southern California Education and Research Center (SCERC) Pilot Project Research Training Program, supported by NIOSH ($10,000, 2026–2027).

Longitudinal Study of Office Workers' Well-being, Performance, and Comfort 2023–Present
Co-Lead Researcher

- Coordinate a 3-year, 60-participant longitudinal field study across five cohorts, integrating environmental sensors (Awair), audio badges, Fitbit wearables, and ecological momentary assessments to capture well-being, performance, and comfort across home and office workspaces
- Apply frequentist and Bayesian mixed-effects models, temporal feature extraction (autocorrelation, entropy, Fano factor), and anomaly detection methods (matrix profiles, seasonal decomposition, LSTM autoencoders) to multimodal longitudinal data
- Led a mediation analysis published in *Building and Environment* showing that office days increase workload demands but improve performance, and that indoor air quality (TVOCs, CO_2, $PM_{2.5}$) partially explains home-office workload differences
- Led a study published in the *Journal of Occupational and Environmental Medicine* (JOEM) linking hybrid workspace location to productivity, mood, and stress, identifying indoor air quality as a candidate occupational health intervention target
- Leading a new analysis of IEQ time-series regime discovery, applying matrix profiles, HMM/ARHMM, LSTM, and change-point detection methods (BOCPD, CPD, BCPD) to identify latent behavioral and environmental states
- Leading an ongoing investigation of privacy-utility trade-offs in behavioral state sequences derived from passive indoor sensing, quantifying re-identification risk and inferential exposure across multimodal data streams
- Mentoring three Ph.D. students (Antonin Brun, Tao Zang, Mireya Reyes) in sensing deployment, data analysis, and manuscript development
- Collaborating on an interdisciplinary team spanning civil engineering (building and indoor environment), psychology and statistical reasoning (Dr. Gale Lucas), occupational science and therapy (Dr. Shawn Roll and lab), and electrical and computer engineering (Dr. Shrikanth Narayanan and lab), integrating environmental, statistical, occupational, and bio-behavioral analytics

Funded by NSF Grant No. IIS-2204942 (2023–2027).

Interactive Effects of IEQ on Cognitive Performance, Comfort, and Stress 2022–2024
Lead Researcher

- Designed a mixed-design controlled experiment with 52 young adults testing interactions among temperature (20°C/25°C), lighting color temperature (2700K/6500K), and noise (50dB/65dB) on cognitive performance, comfort, and stress
- Collected and analyzed physiological, behavioral, and self-report data to evaluate moderating effects of gender and BMI on selective attention, creativity, and stress responses
- Published findings in *Building and Environment* showing that warm, high-noise conditions reduced cognitive performance and comfort, particularly among males and healthy-BMI individuals, while overweight/obese participants showed heightened sensitivity to lighting color temperature
- Published findings in *Indoor and Built Environment* demonstrating that IEQ factors interact with gender and BMI to shape stress responses in complex, person-specific ways, motivating adaptive and demographically personalized environmental design strategies
- Mentored Haotong Fu (B.Sc.) in preprocessing physiological signals, improving the analysis pipeline
- Learned cognitive test embedding methods for human subjects experiments from Dr. Mohamad Awada (then senior Ph.D. student, now Assistant Professor at NYU), and collaborated with Dr. Gale Lucas on study design and proper assignment of dependent, independent, and environmental variables

Funded by NSF Grant Nos. CMMI-1763134 and IIS-2204942.

Energy Efficiency in Affordable Housing Settings 2023–Present
Lead Researcher

- Conducted surveys, interviews, and utility-bill analyses across four affordable housing properties to characterize energy/water use behaviors, energy burden, and conservation motivations among low-income residents
- Published findings from a resident questionnaire across four affordable housing properties in *Energy and Buildings*, identifying satisfaction, energy burden, and conservation drivers, and outlining educational programs and PV solar installation as key intervention strategies alongside implementation barriers
- Published a second study in *Energy and Buildings*, combining in-depth interviews and utility-bill analysis, finding that 82% of residents adopted energy-efficient behaviors to save money while 45% were also environmentally motivated, that PV combined with subsidies reduced bills by up to 80%, and that rigid routines and limited access to monitoring tools were key barriers to demand-response participation
- Leading deployment of low-cost, circuit-level non-intrusive load monitoring (NILM) devices across 107+ units to capture high-resolution energy data and support a cyber-physical system framework for grid-responsive, adaptive housing operations
- Running a resident survey to inform the design of an energy education framework aimed at improving demand responsiveness and lowering energy burden in affordable housing communities
- Mentoring three Ph.D. students (Mireya Reyes, Antonin Brun, Yuxin Yang) on IRB protocols, survey design, and data collection; mentored four Viterbi SHINE high schoolers (Summer 2023) on residential energy dashboard interfaces focused on gamification, sustainability, comfort, and cost transparency
- Collaborating across a multi-institutional partnership, including National CORE (affordable housing developer), ARUP, Studio E Architects, and Built Momentum, with biweekly meetings to track study progress, coordinate resident engagement, identify practice gaps, and develop strategies to reduce energy costs across a diverse portfolio of multifamily, low-income, and permanent supportive housing properties

Funded by California Energy Commission (CEC) Grant No. GFO-20-305.

Equity in the Built Environment 2021–2023
Lead Researcher

- Conducted a systematic review following PRISMA guidelines, screening 6,731 studies down to 232 to map distributional, recognitional, and procedural inequities across U.S. built environments
- Developed the Built Environment–Group–Need (BGN) model, a novel framework integrating physical, social, and procedural dimensions of equity to assess research attention across settings
- Identified major asymmetries in the literature, including neglected settings (commercial, educational, healthcare, energy infrastructure) and underrepresented groups (people with disabilities, neurodivergent individuals, displaced populations, and women)
- Led the literature search and data visualization, and co-authored the manuscript calling for interdisciplinary, community-

based approaches to inclusive design and policy
- Collaborated with Dr. Santina Contreras and Dr. Geoff Boeing, Assistant and Associate Professors of Urban Planning and Spatial Analysis at USC, in defining the study's scope of equity and its underlying dimensions

Taxonomy of Human-Building Interaction (HBI) 2022
Contributing Author

- Investigated the emerging field of Human-Building Interaction (HBI) through ten foundational research questions and expert workshops with 25 academic and industry participants
- Helped identify core HBI principles, trust, control, collaboration, privacy, equity, and sustainability, mapping how human experience and adaptive building systems converge in intelligent environments
- Contributed to manuscript development and editing, and co-authored one subsection establishing conceptual and methodological foundations for the field
- Collaborated with 24 interdisciplinary experts convened through the NSF-funded IN2WIBE network (International Network of Networks in Well-being in the Built Environment), spanning civil and architectural engineering, psychology, occupational science, HVAC and sensing engineering, human-computer interaction, and industry, including Drs. Burcin Becerik-Gerber, Gale Lucas, Shawn Roll, Mario Bergés, Farrokh Jazizadeh, Arsalan Heydarian, and Olga Boric-Lubecke

Funded by NSF Grant No. 2001742.

Resilience Assessment of the Tabriz Historic Bazaar 2021–2022
Research Assistant

- Evaluated the seismic vulnerability and resilience of the Tabriz Historic Bazaar, a UNESCO World Heritage site, using an integrated index-based framework across cultural, historical, physical, socioeconomic, and organizational dimensions
- Applied a multi-criteria analytic hierarchy process (AHP) with fuzzy logic to quantify qualitative parameters across 5 dimensions, 17 indicators, and 50 sub-indicators, identifying priority areas for resilience intervention
- Conducted the GIS-based spatial analysis, developing geospatial layers to map structural vulnerability and resilience patterns across the bazaar's historic urban fabric

Structural Health Monitoring of Civil Infrastructures Using Inverse Eigenvalue Algorithms 2018
M.Sc. Thesis, Sharif University of Technology

- Developed and compared three inverse eigenvalue-based algorithms for structural health monitoring (SHM) and damage detection in civil infrastructures
- Identified stiffness and mass matrices of 2D and 3D shear models using mass modification to detect the location and severity of structural damage
- Extended the framework to incorporate soil-structure interaction effects, enhancing accuracy for realistic conditions such as offshore platforms
- Introduced a novel "double-dimension" formulation enabling identification of lower structural levels from known upper-story parameters and system natural frequencies

---

TEACHING EXPERIENCE

University of Southern California, Los Angeles, USA (2021–Present)
Graduate Teaching Assistant

- **CE 501: Construction Practices** S2023–S2024
  Systems, processes, and constraints in managing major construction projects and delivery [Instructor: Prof. Henry Koffman]. Designed and graded assessments, supervised graders, and held office hours; guided students in connecting planning concepts with real-world scheduling and field applications.

- **CE 469: Sustainable Design & Construction** F2023
  LEED certification, green building strategies, carbon footprinting, and embodied energy calculations [Instructor: Prof. Henry Koffman]. Coordinated guest lectures on sustainability metrics; helped students strengthen LEED exam readiness through applied feedback and targeted quizzes.

- **CE 502: Construction Accounting, Finance, and Strategy** S2022
  Financial analysis, accounting principles, and strategic management frameworks for construction operations [Instructor: Prof. Ed Reynolds]. Led office hours and review sessions, evaluated final presentations, and provided detailed feedback on assignments.
- **CE 334L: Mechanical Behavior of Materials** F2022
  Material science fundamentals, atomic bonding, and mechanical characterization of metals, concrete, wood, composites, and asphalt [Instructor: Prof. Botong Zheng]. Supervised lab sessions and test-result analysis, encouraging students to interpret variability and reflect on experimental uncertainty.
- **CE 573: Advanced Technologies in AEC Practices** F2021
  Data-centric engineering paradigms, digital modeling, and simulation technologies for innovation in AEC projects [Instructor: Prof. David Gerber]. Organized and moderated industry guest lectures and evaluated graduate projects.

Sharif University of Technology, Tehran, Iran (2015–2017)
Teaching Assistant

- **Design of Steel Structures I & II** 2015–2017
  Design principles for steel members and systems, emphasizing load combinations, member behavior, and connection design [Instructor: Prof. Hassan Moghaddam]. Led problem-solving sessions, guided students through code-based calculations, and mentored teams on applying AISC standards.

PANELS, PRESENTATIONS, AND POSTERS

Panels

- **GreenBuild 2024** Nov 2024
  Palm City Transit Village: Reimagining Affordable Housing's Carbon-Constrained Future — represented USC research team alongside panelists from National CORE, ARUP, and Studio E Architects; Philadelphia Convention Center, PA.

Presentations

- **CIB Conference** May 2025
  Occupant Satisfaction with Building Systems and Thermostat Usage in Energy-Efficient Affordable Housing in Southern California: Preliminary Findings — hosted by Purdue University.
- **ASCE i3CE 2025** May 2025
  Longitudinal Analysis of IEQ Variability in Office and Home Workspaces: Insights for the Future of Hybrid Work — hosted by Louisiana State University, New Orleans, LA.
- **ASCE i3CE 2024** Jul 2024
  A Preliminary Investigation for Understanding Temperature's Associations with Office Workers' Psychological Well-Being: A Close Look into Anxiety, Motivation and Concentration — hosted by Carnegie Mellon University, Pittsburgh, PA.
- **UNL Future of Building Industry and PhD Symposium** May 2024
  Interactive Effects of IEQ Factors on Environmental Stress — hosted by Durham School at UNL, Omaha, NE.

Posters

- **NSF CPS Rising Stars Workshop** Mar 2025
  Selected as one of 30 early-career researchers nationwide from 170+ applicants (NSF travel support); Vanderbilt University, Nashville, TN.
- **Southern California CEE Research Symposium** Apr 2025
  A Longitudinal Study of Indoor Environmental Quality in Hybrid Workspaces — hosted by the CEE Departments at USC and UC Irvine.
- **Astani Research Symposium** Sep 2022
  Social Justice in the Built Environment — hosted by the CEE Department at USC.

PROPOSAL WRITING EXPERIENCE

- **SCERC Pilot Project Research Training (PPRT) Program** 2026–2027
  $10,000 — Southern California Education and Research Center (SCERC), supported by NIOSH, through UCLA. PI: Mirmahdi Seyedrezaei; Faculty Sponsor: Dr. Burçin Becerik-Gerber.
  Independently authored and secured funding for *Work Pattern Boundedness and Occupational Stress: A Pilot Study of Environmental and Organizational Drivers in Hybrid Work*, a feasibility study deploying continuous environmental and physiological monitoring across office workers. Led proposal development, study design, and IRB protocol.
- **California Energy Commission (CEC) EPIC Grant, Phase II** 2023–2027
  $8,000,000 — California Energy Commission. PI: National CORE; Research Partner: USC.
  Led the research design, data collection protocols, and analysis plan for Phase 2 deployment across 107 affordable housing units; contributed to proposal development and secured USC's research role, including manuscript preparation and dissemination of findings.
- **NSF Award #2533498: Embodied Intelligence in Smart Workspaces** 2025–2028
  $540,000 — National Science Foundation (CMMI, M3X Program). PI: Dr. Burçin Becerik-Gerber; Co-PIs: Dr. Gale Lucas, Dr. Yasser Khan.
  Contributed to methodology development and proposal writing for this award introducing adaptive human-building collaboration as a new class of intelligent engineered systems, integrating empirical evidence from ongoing longitudinal studies.

MENTORSHIP EXPERIENCE

University of Southern California 2023–Present

Yuxin Yang *(PhD Level)* 2025–Present
Oriented to available datasets and multimodal sensing infrastructure, and guided in defining and refining research questions across the affordable housing energy efficiency project and the BIM-informed building intelligence project. Second-year Ph.D. student at USC.

Tao Zang *(PhD Level)* 2024–Present
Trained in IRB procedures, multimodal data collection, structured data analysis, literature review, and manuscript revision across the longitudinal hybrid work study and worker health feasibility study, developing stronger research planning skills and independence. Third-year Ph.D. student at USC studying Sick Building Syndrome and environmental contributors to occupant well-being.

Mireya Reyes *(PhD Level)* 2024–Present
Supported in developing research fundamentals — IRB training, multimodal data collection, analysis pipelines, literature synthesis, and manuscript revision — across the longitudinal hybrid work study and the affordable housing energy efficiency project, while building prioritization and workflow management skills. Second-year Ph.D. student at USC investigating sleep patterns and workplace environmental factors.

Antonin Brun *(PhD Level)* 2023–Present
Guided through IRB processes, experimental design, multimodal sensing deployments, and data analysis across the longitudinal hybrid work study and the affordable housing energy efficiency project, with continued support in advancing independent research judgment. Fourth-year Ph.D. student at USC, developing human-computer interaction models for smart building controls that enhance occupant health.

Aadish Panchmia *(MSc Level)* Summer 2024
Managed and analyzed household electricity data to support energy equity initiatives; completed an MS in Green Technologies at USC. Now a Project Engineer at Moss & Associates (Construction), Fort Lauderdale, FL.

Dylan Thai *(High School Level, Viterbi SHINE Program)* Summer 2024
Developed predictive models of productivity in workplace environments. Now pursuing a BS in Mechanical Engineering at UC Berkeley.

Viterbi SHINE Program *(High School Level)* Summer 2023

Mentored four students on residential energy dashboard interface design as part of the affordable housing energy efficiency project:

- **Daniel Morales** — gamification/rewarding focus; now studying Civil Engineering at CSU Long Beach.
- **Madeleine Cheung** — environmental sustainability focus; joined NASA JPL, then Civil Engineering at UCLA; co-authored "How a mentoring connection boosted my ambitions for a science career," *Nature* (2025).
- **Moses Ma** — comfort-driven design focus; beginning Urban Design and Architecture Studies at NYU (Fall 2026).
- **Rohin Phukan** — cost-informing design focus; now studying Computer Science at Cornell.

Haotong Fu *(B.Sc. Level)* Spring 2023

Contributed to physiological signal preprocessing as part of the interactive IEQ effects study, improving the analysis pipeline. Now pursuing an MS in Computer Science at USC.

AWARDS AND HONORS

Scholarships / Fellowships

- USC Graduate School Fellowship 2024–2025

Awards

- USC CEE Outstanding Research Assistant of the Year 2025
- Top Peer Reviewer, *ASCE OPEN* 2025
- Das Family Travel Award 2023, 2024, 2025
- First place, Human Building Interaction Online Hackathon (NSF-funded) 2022

Qualifications & Rankings

- Qualified member, Iranian National Elites Foundation 2016–Present
- Exceptional talent admission to M.Sc. in Earthquake Engineering 2016
- Top 10% of B.Sc. Civil Engineering class of 2012 2016
- Top 0.1% among 260,000 participants, Iranian nationwide entrance exam 2012

SERVICE

Reviewer for Peer-Reviewed Journals 2023–Present

- *Building and Environment*; *Energy and Buildings*; *Journal of Building Engineering*; *Cities*; *Journal of Environmental Psychology*; *Utilities Policy*; *Energy Policy*; *Journal of Housing and the Built Environment*; *Perspectives in Architecture and Urbanism* 2025–Present
- *Journal of Computing in Civil Engineering* 2024–Present
- *ASCE OPEN: Multidisciplinary Journal of Civil Engineering* 2023–Present

Reviewer for Peer-Reviewed Conferences 2024–Present

- *ASCE International Conference on Computing in Civil Engineering (i3CE)* 2024, 2025

Other Service

- **Member**, ASCE Visualization, Information Modeling, and Simulation Committee; CIB W107 Construction in Developing Countries Committee 2025
- **Conference Executive Manager**, Iran Nationwide Flood Conference, SOSA Poverty, Sharif University of Technology 2019
- **Contributor**, Iran National Flood Report — proposed a new approach to post-disaster relief activities, termed *"Green Relief"* 2019

INDUSTRY EXPERIENCE

- **Structural Retrofit Designer and Researcher** 2018–2019
  Sharif Residential Structural Services Research Group (HED Research Group), Tehran, Iran

- Contributed to the safety and resilience of aging residential infrastructure through seismic retrofit design and planning.
- Conducted structural assessments and developed retrofit strategies for residential buildings.
- Collaborated with faculty researchers and practitioners to bridge research outcomes with industry applications.

- **Intern Assistant Engineer** Summer 2014 & 2015
Atigh Commercial Complex Project (supervised by Prof. Mohammad Mehdi Mortaheb), Tehran, Iran
- Gained hands-on experience in large-scale commercial construction, assisting with site operations, logistics, and quality control.
- Supported quantity surveying, cost estimation, and procurement planning.
- Monitored the integration of structural design and construction management on the project.

MEDIA COVERAGE

- **"Stressed at work? USC Researchers Aim to Change That"**, USC Viterbi School of Engineering (Mar. 2023). Reports on an interdisciplinary USC team awarded an NSF SCH grant to mitigate job stress. Link
- **"Mitigating On-the-Job Stress"**, USC Chan Magazine (Fall 2023/Winter 2024). Describes the NSF-funded USC project using AI and sensor systems to detect and counteract workplace stress. Link
- **"Against the Grain: Research Reinforces Message That Wood Is Good"**, WorkTech Academy (2025). Highlights research on the environmental and structural benefits of wood construction. Link
- **"Palm City Transit Village, Zero Emission Affordable Housing Design"**, California Energy Commission (2024). Presents the design study for a 288-unit all-electric, transit-oriented affordable housing development. Link
- **"Palm City Transit Village: Reimagining Affordable Housing's Carbon-Constrained Future"**, Streamly Video (Nov. 2024). Recorded at GreenBuild 2024; narrates the project and its low-emission, smart-infrastructure vision. Link
- **"How a Mentoring Connection Boosted My Ambitions for a Science Career"**, *Nature* (Sep. 2025). Reflects on how USC's SHINE program fostered a high-school student's engagement in scientific research. Link

VOLUNTEER EXPERIENCE

- **SOSA Poverty** — Social-Environmental Activist & Volunteer Member (2014–2022); Voluntary Leadership and Team-Building (2017–2021) 2014–2022
Contributed to an international NGO with general consultative status at the UN ECOSOC, engaging in grassroots initiatives addressing poverty and environmental challenges in Iran and abroad. Built and led multidisciplinary volunteer teams in humanitarian response, knowledge management, and educational planning, including organizing the NGO's response teams during the 2019 Iran nationwide floods.

REFERENCES

All references are affiliated with the University of Southern California, USA.

- **Burçin Becerik-Gerber** — Professor, Sonny Astani Dept. of Civil and Environmental Engineering. becerik@usc.edu
- **Gale Lucas** — Research Associate Professor, Institute for Creative Technologies; Depts. of Computer Science, Civil and Environmental Engineering, and Psychology. lucas@ict.usc.edu
- **Shawn Roll** — Professor, USC Chan Division of Occupational Science and Occupational Therapy. sroll@usc.edu
- **Lucio Soibelman** — Professor, Sonny Astani Dept. of Civil and Environmental Engineering. soibelma@usc.edu
- **Shrikanth (Shri) Narayanan** — Professor, Depts. of Electrical and Computer Engineering, Computer Science, Linguistics, Psychology, and Neuroscience. shri@usc.edu